

The CSS Box Model

Understand how content, padding, borders, and margins determine the space an element occupies.

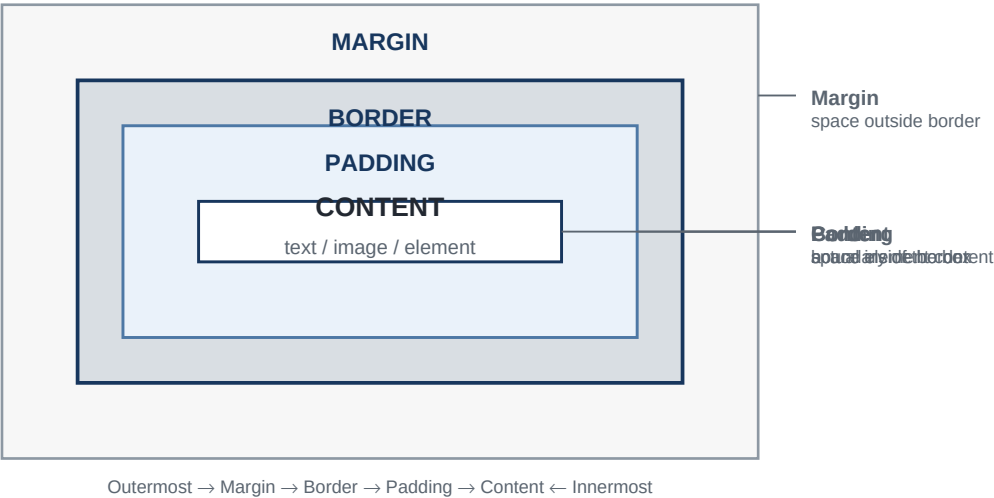
■ What is the Box Model?

Every HTML element on a page is treated as a rectangular box in the browser. The CSS Box Model defines how that box behaves and is the foundation of spacing, layout, and sizing in CSS.

The Four Parts

Layer	What it represents
1. Content	The actual text, image, or other content inside the box.
2. Padding	Space inside the box, between the content and the border.
3. Border	The boundary surrounding the padding and content.
4. Margin	Space outside the border, used to create distance between elements.

■ Box Model Structure



1. Content

The content is the innermost part of the box. It contains the actual text, image, or other element content.

2. Padding

Padding is the space inside the box, between the content and the border. Increasing padding pushes the content inward and increases the total size when using the default **content-box** sizing model.

3. Border

The border surrounds the padding and content. You can control its width, style, and color.

```
width: 200px;
height: 100px;
padding: 20px;
border: 2px solid black;
```

```
margin: 30px;
```

4. Margin

Margin is the space outside the border. It is commonly used to create distance between elements. Margins do not have a background color and are transparent.

■ How the Box Size Is Calculated

With the default **content-box** model, the declared width and height apply to the content. Padding and borders are then added around it.

Dimension	Calculation	Result
Width	$300 + 2 \times 20 \text{ padding} + 2 \times 5 \text{ border}$	350px
Height	$150 + 2 \times 20 \text{ padding} + 2 \times 5 \text{ border}$	200px

Margin is outside this box, so it adds extra space between elements rather than becoming part of the element's declared content-box dimensions.

Box Sizing: content-box vs border-box

CSS provides the **box-sizing** property to control how an element's declared width and height are calculated.

Mode	How width and height work
content-box	The specified width and height apply to the content only. Padding and border are added outside those dimensions.
border-box	The specified width and height include the content, padding, and border. The total size stays fixed.

Default: content-box

By default, CSS uses **content-box**. For example:

```
.box {  
  width: 300px;  
  height: 150px;  
  padding: 20px;  
  border: 5px solid gray;  
  margin: 40px;  
}
```

Using border-box

With **border-box**, the specified width and height remain fixed. Padding and border are included inside those dimensions.

```
* {  
  box-sizing: border-box;  
}
```

A common layout pattern is to apply border-box globally so width and height calculations are easier to reason about.

Visual Example

```
.card {  
  width: 400px;  
  padding: 20px;  
  border: 10px solid black;  
  box-sizing: border-box;  
}
```

In this example, the total width of the element remains 400px, including its padding and border.

Quick Reference

Part / Property	Key idea
Content	The actual text, image, or element inside the box.
Padding	Space between content and border.
Border	Boundary around padding and content.
Margin	Space outside the border.
content-box	Width and height apply to content; padding and border are added.
border-box	Width and height include content, padding, and border.
box-sizing	Controls how declared width and height are calculated.

■ Tip

- Understand how content, padding, border, and margin interact before adjusting spacing.
- Remember that margin sits outside the border.
- Use **box-sizing: border-box** when you want declared dimensions to include padding and borders.

Quick Takeaway

The CSS Box Model controls how every element takes up space. Understanding the relationship between content, padding, border, and margin makes CSS sizing and layout much easier to manage.